

Certificate

This is to certify that the Project/ Dissertation entitled, CBSE Database Management Software, is a bonafide work done by Master/Miss Jaiditya Ranjan of Class 12 - 'B', Session 2025-26, in partial fulfillment of Session Ending Exam 2026 and has been carried out under my direct supervision and guidance.

This report or a similar report on the topic has not been submitted for any other examination and does not form a part of any other course undergone by the candidate.

Acknowledgment

As usual, a large number of people deserve my thanks for the hel they provided to me for the preparation of this project.

First of all, I would like to thank my teacher PGT Dinesh Kumar for his support during the preparation of this project. I am very thankful for his guidance.

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Objective

The primary objective of this project, titled CBSE Database Management Software, is to mimic Central Board of Secondary Education's(henceforth referred to as 'CBSE') main website, <https://cbse.gov.in> to:

- facilitate parents to see schools that are affiliated with CBSE,
- for candidate's to view their Exam Admit Card,
- and for students to recieve the final marksheet for the successfull completion of the Board exams by students of class 10(Secondary School Examination) and class 12(Senior Secondary School Examination).

For this purpose, we have chosen MySQL as the back-end server to handle the mock CBSE-database, Python-Tkinter as the front-end Graphical User Interface(henceforth referred to as 'GUI'), and Python as the connector between the front-end and the back-end.

The project also seeks to demonstrate fundamental programming concepts such as modular programming and Python-MySQL connectivity, aligning with the foundation knowledge expected from CBSE's (83)Computer Science course for class 12.

Scope of the Project

What the project aims to cover:

- To mimic the official CBSE website.
- For the data of affiliated schools, we found CBSE's official website:
<https://saras.cbse.gov.in/SARAS/AffiliatedList/ListOfSchdirReport> which provided us details of all affiliated school. We decided to get the data manually from here.
- Since for the past 3 years, CBSE has reported about 12-13 lakhs registered students for the Boards exams, we have

What the project does not aims to cover:

- Rest of CBSE's website, as it falls out of our scope, and also we are very limited in our knowledge of how to implement them.

For the project, the following general assumptions were made and followed throughout the entire codebase:

- **Database:** We have assumed that with the start of every academic year, CBSE makes a new database named CBSE_EXAM_RESULT_YXXXX, where XXXX refers to the present year. Following this assumption, we have assumed that CBSE makes 4 tables in the database:
 - **COURSES:** courses offered by CBSE for the present academic year,
 - **AFFILIATED_SCHOOLS:** schools that are affiliated with CBSE,
 - **REGISTERED_STUDENTS:** students that are registered for the academic year,
 - and **EXAM_RESULTS:** stores results of all students.
- **Number of students:** Every year, CBSE has reported a surge in number of students who have appeared for the Board exams^{[17][17]}. In the year 2025 alone, approximately 17.8 lakhs and 24.2 lakhs students appeared for the Senior Secondary School and Secondary School examination, respectively. However, handling such huge amount of student count is not feasible for us. As such, we decided that 5 lakhs students will be appearing for the Board exam, where the number of class 12 and class 10 students are in the ratio 3:2.
- **No absentees:** Another assumption has been made that all appearing students will be present in all of their

examinations, i.e. no one will be absent for any of their examinations.

- **Subject count for each student:** Following the latest curriculum guidelines for the academic year 2025-26 published by CBSE for both class 10 and class 12 students, all students must have 5 mains subjects and 1 additional subject mandatorily, and then it is upon the student to choose if they wish to have more subjects. However, we have decided that each and every student will get a total of 7 subjects, where 5 subjects are the main subjects and 2 are additional subjects.

During the project, the following limitations were faced due to the added complexity of handling Python and MySQL simultaneously:

- **Limited number of school data:** Since we had to manually get the details for all affiliated schools from a website^[17], and also take a screenshot of each school's data, we could only have 107 schools in our database, which greatly increases the number of students present in each school. From our test runs, we found that all schools have roughly 4000-5000 students, which is highly unrealistic.
- **Student being assigned exam center outside of their district:** We also wanted to ensure that each students get assigned a exam centre in their own district, but we have failed to do so due to complexity.
- **Subject count for each student:** All students are expected to have 5 main subjects and 2 additional subjects. However, in real life, a student can have 5 main subjects, and 1 additional subject that they have to choose, and then any other subject of their choice if they wish to take more subjects. So, students can have 6, 7, or even 9 subjects if they wish to. However, we have had to hard-code 7 subjects for all students. Another aspect that we could not workout on was that according to the CBSE curriculum, a class 10 student is expected to have at least 2 main subjects as language subjects L1 and L2, and in their additional subjects they could choose from any subject language or otherwise, and a class 12 student is expected to have only 1 main subject as language subject L1, and then in their additional subjects they can choose from language subjects or otherwise. But, again citing the complexity of our project, we could not code this separate behaviour, and as such a student could have all

5 subjects as language subjects(theoretically), or have no language subjects at all.

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Theoretical Background

The project leverages key theoretical concepts and principles from Computer Science and SQL to build a GUI-based ...

System Planning

The development of the CBSE DBMS was systematically planned and executed in the following phases:

1. Requirements Analysis

- Identify the key functionalities to be included in the software.
- Determine the user requirements for a simple and intuitive interface.

2. Design

- Develop a modular design by creating separate Python modules for each functionality.
- Design a main program to serve as the entry, providing a GUI-based interface to use the software.

3. Implementation

- Write the code for each module, focusing on accuracy, efficiency, and error handling.
- Implement the main program to integrate the modules and provide seamless experience.

4. Testing

- Test each module independently to ensure its functionality.
- Conduct integration testing to verify that the modules work together as intended.
- Perform user testing to identify and resolve any usability issues.

5. Documentation

- Document the software's purpose, functionality, and structure.
- Include clear instructions for users and developers, ensuring the project is easy to understand and extend.

6. ...

Methodology Adopted and Details of Hardware/ Software Used

For this project, we developed a GUI-based CBSE DBMS using MySQL as the backend software, Tkinter for the GUI, and Python to integrate everything in one place. To connect to the MySQL server/database, we designed a module named 'connection_module.py', and to design the GUI, we developed a module named 'tkinter_gui.py'.

- **Hardware:** The software was developed and tested on a personal P.C. with standard specifications: Dual-boot operating system with Linux Mint v22.2 and Windows 10 64-bit, 4 GB DDR3 RAM, 512 GB HDD ROM, and an Intel Core i5-3570k processor with built-in Intel HD Graphics 4000.
- **Software:** Python 3.14, MySQL v8.0.43-0ubuntu0.24.04.2 for Linux on x86_64 (Ubuntu), Tkinter vX.XX, and the software was developed and tested on PyCharm as the Integrated Development Environment(IDE).

Snippet of Source Code

Snippets of Test Run

User Manual

Brief Background of Organization

Reference

1. CBSE main website: <https://www.cbse.gov.in>
2. List of affiliated schools: <https://saras.cbse.gov.in/SARAS/AffiliatedList/ListOfSchdirReport>
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5. Year-wise summary of number of students who appeared for the Class 12 exams: <https://www.timesnownews.com/education/cbse-board-results-2025-number-of-students-who-appeared-for-cbse-class-12-exam-since-2017-article-151431303>
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